An Efficient Cross-Modality Self-Calibrated Network for Hyperspectral and Multispectral Image Fusion

Huapeng Wu, Jie Gui, *Senior Member, IEEE*, Yang Xu, *Member, IEEE*, Zebin Wu, *Senior Member, IEEE*, Yuan Yan Tang, *Life Fellow, IEEE*, and Zhihui Wei, *Member, IEEE*

***Abstract*— Recently, deep convolutional neural network (CNN)-based hyperspectral and multispectral image fusion methods have shown significant performance. Nevertheless, the rich spatial and spectral details of hyperspectral images (HSIs) have not been fully explored, leaving room for further improve the representation ability of the model. In this article, we propose an efficient cross-modality self-calibrated network (CMSCN) for hyperspectral and multispectral image fusion. Specifically, we use a cross-modality nonlocal module (NL) to fuse a high-resolution multispectral image (HR-MSI) and a low-resolution hyperspectral image (LR-HSI) to get an enhanced LR-HSI. In addition, a novel cross-scale self-calibrated convolution structure is proposed to explore and exploit multiscale and hierarchical spatial-spectral features, which can improve the learning ability of the model. The introduced efficient spatial-spectral attention mechanism can calibrate the feature representation at different dimensions, thereby providing more efficient and accurate information for HSI reconstruction. Extensive experimental results on various HSIs demonstrate the superiority of our method in comparison with the state-of-the-art image fusion methods.**

***Index Terms*— Attention mechanism, convolutional neural network (CNN), hyperspectral and multispectral image fusion, multiscale features.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain rich spatial and spectral information, which are widely used in various remote sensing fields, such as mineral classification [1], [2] and detection [3], [4]. However, due to the hardware limitations, the acquired HSI is often low resolution, which severely limits subsequent high-level vision tasks. Hyperspectral image super-resolution (SR) is an image post-processing techniques that can infer a high-resolution hyperspectral image (HR-HSI) from degraded low-resolution images [5]. According to the number of input degraded images, hyperspectral image SR methods can be divided into two categories: single hyperspectral image SR [6], [7] and fusion-based hyperspectral image SR [8], [9].

Compared with single hyperspectral image SR, fusion-based hyperspectral image SR tries to reconstruct the SR hyperspectral image (SR-HSI) by merging the low-resolution hyperspectral image (LR-HSI) with the high-resolution multispectral image (HR-MSI). Recently, thanks to the powerful learning ability of deep learning, nature image SR methods based on deep convolutional neural network (CNN) have achieved great performance. Its main principle is to learn the mapping function from LR patches to their HR counterparts. Dong et al. [10] first present a three-layer super-resolution convolutional neural network (SRCNN) for a single-image SR. Later, other representative methods are proposed by integrating residual learning, dense connections, and recursive learning, including deeply-recursive convolutional network (DRCN) [11], enhanced deep super-resolution (EDSR) [12], residual dense network (RDN) [13], etc. Inspired by superior performance in nature image SR, some remote sensing image fusion methods based on deep learning are proposed. In [14], the upsampled LR-HSI and the HR panchromatic images are concatenated as the input of the network to learn the target HR-HSI. Palsson et al. [15] utilize a 3-D CNN to fuse a LR-HSI and a HR-MSI. Dian et al. [16] propose a deep HSI sharpening method that incorporates the image priors learned by residual learning-based CNN into the fusion framework. Yuan et al. [17] propose a multiscale and multidepth CNN (MSDCNN)-based pan-sharpening. Shao and Cai [18] apply a two branches network to learn spectral and spatial feature information of the multispectral image and panchromatic image separately. Xie et al. [19] introduce a model-based deep learning method for HSI/MSI fusion. Instead of using a traditional iterative algorithm, this method designs a deep learning-based proximal operator to solve the model and obtains the state-of-the-art HSI SR results. Hu et al. [20] propose a novel HSI/MSI fusion network (HSRNet), which mainly contains two parts: one is used to preserve the spectral information and the other aims to get satisfied spatial details.

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Huapeng Wu is with the School of Computer Science, Nanjing Audit University, Nanjing 211815, China.

Jie Gui is with the School of Cyber Science and Engineering, Southeast University, Nanjing 211100, China.

Yang Xu and Zebin Wu are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China.

Yuan Yan Tang is with the Zhuhai UM Science and Technology Research Institute, Faculty of Science and Technology, University of Macau, Macau, China.

Zhihui Wei is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China, and also with the School of Science, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: gswei@njust.edu.cn).

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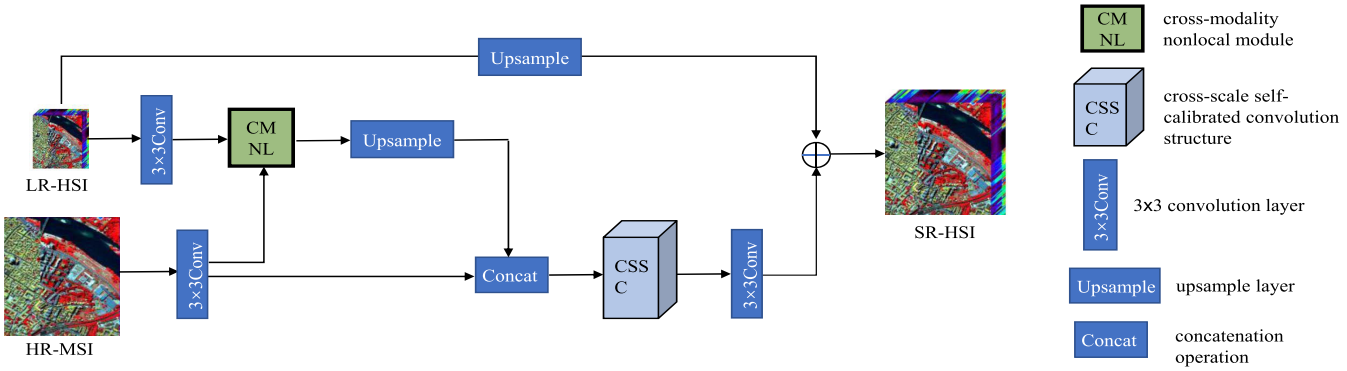


Fig. 1. Architecture of our CMSCN. ⊕ denotes broadcast element-wise addition.

Although the above methods achieve good results, leaving room for further improves the performance of the network. For example, the upsampled LR-HSI and HR-MSI are often directly concatenated as the input of the network, which will lead to some spatial-spectral information loss. Considering that the HR-MSI contains rich high-resolution (HR) spatial content and the LR-HSI holds most of the spectral information, we propose to use a cross-modality nonlocal module (CMNL) to extract HR spatial details from the HR-MSI and inject them into the LR-HSI. The proposed module can effectively learn the HR-MSIs spatial information with a few computational burden, and preserving the spectral information by a local residual connection [21], [22]. To further explore and exploit more spatial-spectral feature information of HSI, we design a novel cross-scale encoder–decoder network with self-calibrated convolution. Unlike traditional encoder–decoder structure [23], [24] only bridging the information flow between low-level features and corresponding high-level features at same scale, the proposed network can fuse multiscale features of the encoder part and improve the cross-scale information propagation in the encoder-decoder architecture, which will improve the feature representation ability of the model. Specifically, in encoder–decoder structure, we adopt self-calibrated convolution with an efficient spatial-spectral attention (ESSA) layer instead of traditional residual convolution structure as the basic unit, which can efficiently calibrate the feature response and boost the information flow [24], [25], [26]. Moreover, the introduced ESSA can effectively integrate positional information into channel attention (CA) to enable the model to capture channel-wise dependencies and spatial information simultaneously, which is beneficial to enhance feature representation and avoid the loss of spatial-spectral information.

Overall, in this article, we propose an efficient cross-modality self-calibrated convolution network (CMSCN) for hyperspectral and multispectral image fusion. As shown in Fig. 1, the input LR-HSI is first enhanced by learning the long-range interdependency between the LR-HSI and the HR-MSI. The enhancement value of each pixel of the LR-HSI is the weighted sum of all pixels of the HR-MSI according to the learned pairwise correlation (illustrated in Fig. 2). Subsequently, the output features are cascaded with the HR-MSI features to form a feature cube as the input of the proposed cross-scale self-calibrated convolution structure to further learn informative spatial-spectral feature information. Extensive experiments on different hyperspectral datasets show the superiority of the proposed method in comparison with the state-of-the-art image fusion methods.

In summary, the two main contributions of this work are as follows.

1) We introduce the CMNL to hyperspectral and multispectral image fusion and show how to extract HR spatial details of the HR-MSI injecting them into the LR-HSI. The proposed module can effectively enhance feature representations and improve image fusion performance.

2) We further introduce the cross-scale self-calibrated convolution structure with the ESSA to flexibly exploit rich multiscale spatial-spectral features and enhance feature representation ability of the network. Extensive experimental results demonstrate the superiority of our method in comparison with the state-of-the-art HSI fusion methods.

The rest of this work is organized as follows. Section II, we analyze the proposed CMSCN in detail. Experimental results and analysis are given in Section III. Finally, conclusions are reported in Section IV.

II. Proposed Method

A. Network Architecture

In Fig. 1, we show the overall framework of the proposed CMSCN. It mainly consists of two parts, i.e., feature extraction part and global residual connection part. Specifically, $X^L \in \mathbb{R}^{C\times h\times w}$ denotes the input LR-HSI that has the same number of spectral bands as the corresponding ground truth hyperspectral image $Z \in \mathbb{R}^{C\times H\times W}$. $Y^H \in \mathbb{R}^{c\times H\times W}$ denotes the input HR-MSI. $O \in \mathbb{R}^{C\times H\times W}$ is the output super-resolution hyperspectral image (SR-HSI). Our goal is to predict the HR-HSI O from the input LR-HSI X^L and HR-MSI Y^H by training the proposed image reconstruction network

$$O = F_{\text{CMSCN}}\left(X^L, Y^H\right) \tag{1}$$

where F_{CMSCN} is the function of the proposed CMSCN.

The proposed network is optimized with the L_1 loss function by minimizing the difference between the output

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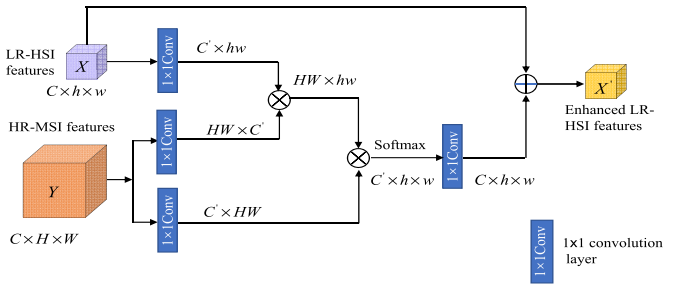


Fig. 2. Structure of CMNL. ⊗ denotes matrix multiplication. ⊕ denotes broadcast element-wise addition.

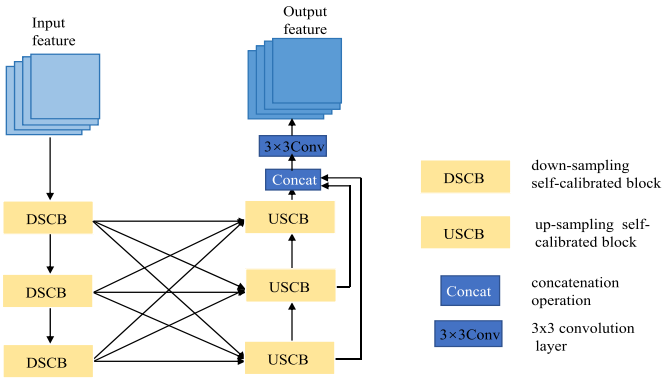


Fig. 3. Flowchart of CSSC.

image O and the ground-truth HR-HSI Z. The loss function is formulated as

$$L(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left|F_{\mathrm{CMSCN}}\left(X_i^L, Y_i^H\right) - Z_i\right|_1 \tag{2}$$

where N denotes the number of training sets and θ is the parameter set of the proposed network. In Section II-B, we will introduce the proposed two basic architecture in detail, including CMNL, cross-scale self-calibrated convolution structure.

B. Cross-Modality Nonlocal Module

To improve the representation ability of the network, the NL is often introduced to learn long-rang spatial information. Let $X = \{x_i\}_{i=1}^{N_L}$ denotes the input LR-HSI features that has C feature maps with size of $h \times w$, and $N_L = h \times w$. The NL can be defined as

$$x_i' = \frac{1}{C(x)}\sum_{j=1}^{N_L}\left(M(x_i, x_j)\cdot g\left(x_j\right)\right) + x_i \tag{3}$$

where x_i' is the output feature. $M\left(x_i, x_j\right)$ denotes the similarity relationship between the ith position and the jth position. $C(x)$ is a normalization factor. $g(\cdot)$ is a linear embedding function (e.g., convolution with kernel size).

However, considering that high-resolution spatial information mainly contained in the HR-MSI, LR-HSI has most spectral information and lacks high-resolution spatial information. In this article, we introduce a CMNL between the LR-HSI and the HR-MSI, which aims to capture rich spatial contextual contents in HR-MSI and inject them into the LR-HSI. Thanks to this module, the proposed network can not only effectively obtain the missing spatial information of the LR-HSI, but also perfectly preserve the spectral contents. As illustrated in Fig. 2, $X = \{x_i\}_{i=1}^{N_L}$ denotes the input LR-HSI features that has C feature maps with size of $h \times w$, and $N_L = h \times w$. $Y = \{y_i\}_{i=1}^{N_H}$ denotes the input HR-MSI features that has C feature maps with size of $H \times W$, and $N_H = H \times W$. First, the pairwise correlation between LR-HSI′s pixels and HR-MSI′s pixels is measured by

$$S_{ij} = M\left(\theta(x_i)^T, \varphi\left(y_j\right)\right) = \frac{\exp\left(\theta(x_i)^T\varphi\left(y_j\right)\right)}{\sum_{j=1}^{N_H}\exp\left(\theta(x_i)^T\varphi\left(y_j\right)\right)} \tag{4}$$

where $x_i \in \mathbb{R}^{C\times 1\times 1}$, $y_j \in \mathbb{R}^{C\times 1\times 1}$, $S \in \mathbb{R}^{hw\times HW}$, $\theta(\cdot)$ and $\varphi(\cdot)$ are two linear embedding functions (e.g., convolution with kernel size 1×1). S_{ij} denotes the similarity relationship between the ith pixel of the LR-HSI and the jth pixel of the HR-MSI. In this article, the measure operation $M(\cdot,\cdot)$ is implemented using a softmax function. Then, the final enhanced output x_i' at the LR-HSI′s ith pixel is the weighted average of all pixels in the HR-MSI according to the learned pairwise correlation

$$x_i' = \sum_{j=1}^{N_H}\left(S_{ij}\cdot g\left(y_j\right)\right) + x_i \tag{5}$$

where j is the index that enumerates all locations in the HR-MSI, capturing rich high-resolution spatial information. $\cdot$ is element-wise multiplication. $+x_i$ represents the residual skip connection that guarantee original spatial-spectral feature information.

C. Cross-Scale Self-Calibrated Convolution Structure

To further learn more informative spatial-spectral features, we design a cross-scale self-calibrated convolution structure with the ESSA instead of traditional residual structure, which enables the model to adaptively learn multiscale feature information during the encoding and decoding process. As shown in Fig. 3, the proposed method adopts down-sampling self-calibrated block (DSCB) and up-sampling self-calibrated block (USCB) as the basic blocks of feature encoder and decoder to learn and calibrate feature representation. In addition, considering that multiscale and multilevel features can provide rich complementary information, we introduce cross-scale connections (CSs) in the down and up-sampling projection units, yielding cross-scale self-calibrated convolution structure. More specifically, each self-calibrated convolution block mainly contains three branches: the first two branches are responsible for recalibrating the feature response and enables the model to focus on important feature information using the ESSA, the other one aims to guarantee the flow of the original information.

1) Down-Sampling Self-Calibrated Block (DSCB): Fig. 4(a) shows the structure of the proposed DSCB, which aims to map the input HR features to the low-resolution (LR) features. Specifically, the input HR feature L^{i-1} is first mapped to an intermediate LR feature map L_{lr}^{i}, then map it back to HR feature map L_{hr}^{i}. Another branch generates attention weights W_{ESSA} on the input feature L^{i-1} through the above ESSA mechanism. Subsequently, the obtained attention weights are

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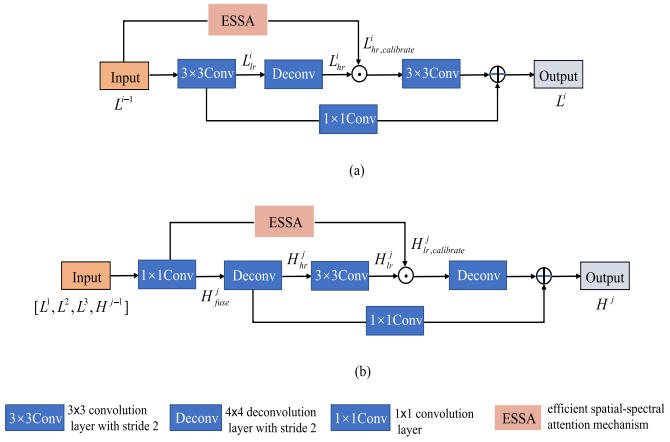


Fig. 4. Flowchart of (a) DSCB and (b) USCB. $\odot$ denotes element-wise multiplication. $\oplus$ denotes broadcast element-wise addition.

multiplied by L^i_{hr} to get the calibrated feature $L^i_{hr,\text{calibrate}}$. Then, the calibrated feature $L^i_{hr,\text{calibrate}}$ is again mapped to LR feature space and added to L^i_{lr} to get the output feature L^i. The down-sampling block can be presented as follows.

The scale down stage

$$L^i_{lr} = W_{\text{Conv}} * L^{i-1} \downarrow . \tag{6}$$

The scale up stage

$$L^i_{hr} = W_{\text{Deconv}} * L^i_{lr} \uparrow . \tag{7}$$

Calibration feature

$$L^i_{hr,\text{calibrate}} = L^i_{hr} \cdot \text{ESSA}\left(L^{i-1}\right). \tag{8}$$

Output feature map

$$L^i = W_{\text{Conv}} * L^i_{hr,\text{calibrate}} \downarrow + W_{1\times1} * L^i_{lr} \tag{9}$$

where W_{Conv} is a 3×3 convolution layer with stride of 2, W_{Deconv} is a 4×4 deconvolution layer with stride of 2. ESSA$(\cdot)$ denotes the ESSA mechanism, $W_{1\times1}$ is a 1×1 convolution layer. $*$ and $\cdot$ are convolution operation and element-wise multiplication, respectively. $\downarrow$ and $\uparrow$ denote the down- and up-sampling operation, respectively.

2) Up-Sampling Self-Calibrated Block (USCB): To improve information exchange among different branches, we construct CSs in the encoder–decoder structure. Specifically, as shown in Fig. 4(b), in the jth USCB, the input features are the concatenation of the output features from all DSCBs and $j-1$th USCB. The process of USCB is similar to DSCB.

Cross-scale feature fusion

$$H^j_{\text{fuse}} = W_{\text{fuse}} * \left[L^1, L^2, L^3, H^{j-1}\right]. \tag{10}$$

The scale up stage

$$H^j_{hr} = W_{\text{Deconv}} * H^j_{fuse} \uparrow . \tag{11}$$

The scale down stage

$$H^j_{lr} = W_{\text{Conv}} * H^j_{hr} \downarrow . \tag{12}$$

Calibration feature

$$H^j_{lr,\text{calibrate}} = H^j_{lr} \cdot \text{ESSA}(H^j_{\text{fuse}}). \tag{13}$$

Output feature map

$$H^j = W_{\text{Deconv}} * H^j_{lr,\text{calibrate}} \uparrow + W_{1\times1} * H^j_{hr} \tag{14}$$

where $[\cdot]$ is the feature concatenation operation. W_{fuse} denotes a 1×1 convolution layer to reduce the channel dimensionality. $j \geq 1$, $H^0 = 0$.

To further enrich spatial-spectral feature information, we use a hierarchical feature fusion strategy to fuse the multiscale feature information in the decoding part. The final output of the encoder and decoder structure is

$$F_{\text{output}} = W_{3\times3} * \left[W_{4up}\left(H^1\right), W_{2up}\left(H^2\right), H^3\right] \tag{15}$$

where $W_{4up}(\cdot)$ and $W_{2up}(\cdot)$ denote the up-sampling operator with scaling factor 4 and 2, respectively. $W_{3\times3}$ is a 3×3 convolution layer to reduce the feature channels, thereby adding with the up-sampled LR-HSI to generate the desired hyperspectral SR image.

3) ESSA Mechanism: Considering that spatial and spectral features are crucial for HSI SR, many articles employ the spatial and CA mechanism to learn spatial and spectral relationships to enhance the representation ability of the network [20], [26], [27], [28]. squeeze-and-excitation network (SENet) [27] introduces CA to recalibrate the feature responses in a channel-wise manner. Given the input feature $X \in \mathbb{R}^{C\times H\times W}$, the CA in SENet can be expressed as

$$w = \sigma\left(W_2\delta(W_1 g(X))\right) \tag{16}$$

where w denotes the output weights, $\sigma(\cdot)$ and $\delta(\cdot)$ are the sigmoid function and rectified linear units (ReLU) activation function. $W_1 \in \mathbb{R}^{C/r\times C\times1\times1}$, $W_2 \in \mathbb{R}^{C\times C/r\times1\times1}$, r is the reduction radio. $g(X) = (1/H\times W)\sum_{i=1}^{H}\sum_{j=1}^{W} X_{ij}$ is the global average pooling operation. Later, in convolutional block attention module (CBAM) [28], the CA and the spatial attention are simultaneously used to recalibrate the feature. The CA in CBAM can be written as

$$w_c = \sigma\left(W_2\delta(W_1 g(X)) + W_2\delta(W_1 h(X))\right) \tag{17}$$

where w_c denotes the output channel weights. $W_1 \in \mathbb{R}^{C/r\times C\times1\times1}$ and $W_2 \in \mathbb{R}^{C\times C/r\times1\times1}$ are shared for $g(X)$ and $h(X)$. $h(X) = \max_{H,W}(X)$ is the global max pooling operation, respectively. The spatial attention in CBAM can be written as

$$w_s = \sigma\left(W^{k\times k}\left[A^s_{\text{avg}}, A^s_{\text{max}}\right]\right) \tag{18}$$

where w_s denotes the output spatial weights and $W^{k\times k}$ is a convolution layer with size of $k\times k$. $A^s_{\text{avg}} \in \mathbb{R}^{1\times H\times W}$ and $A^s_{\text{max}} \in \mathbb{R}^{1\times H\times W}$ are generated using the average pooling operation and the max pooling operation along the channel axis, respectively. $[\cdot,\cdot]$ denotes the concatenation operation.

However, SENet only consider the interdependencies among channels. Although CBAM introduces spatial attention via convolution operation, convolution operation can only learn local spatial information. In this article, we introduce an ESSA to capture the channel relationships and aggregate position-sensitive information along the two spatial directions [26]. Specifically, as shown in Fig. 5, $X_a \in \mathbb{R}^{C\times H\times W}$ denotes the input features, where H, W, and C are height,

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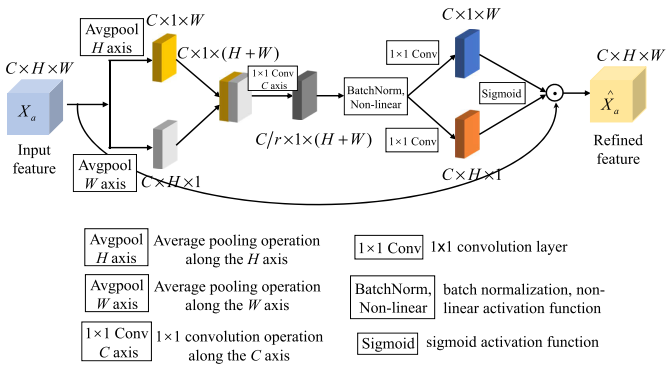


Fig. 5. Flowchart of ESSA. ⊙ denotes element-wise multiplication.

width, and the number of channels, respectively. We first encode channel-wise statistics to produce two 2-D feature maps ($X_a^H \in \mathbb{R}^{C\times 1\times W}$, $X_a^W \in \mathbb{R}^{C\times H\times 1}$) by the 1-D average pooling operation along the H axis and the W axis, respectively. The produced two feature maps can learn long-range information along the height and width directions, which helps the model to augment the spatial-spectral feature representations. Then, they are concatenated and passed into three 1×1 convolutional layers to learn channel-wise interdependencies with precise positional information, which can be written as

$$F = \delta\left(W_1\left[X_a^H, X_a^W\right]\right) \tag{19}$$

$$\alpha^H = \sigma\left(W_2\left(F^H\right)\right) \tag{20}$$

$$\alpha^W = \sigma\left(W_3\left(F^W\right)\right) \tag{21}$$

where $W_1 \in \mathbb{R}^{C/r\times C\times 1\times 1}$, $W_2 \in \mathbb{R}^{C\times C/r\times 1\times 1}$ and $W_3 \in \mathbb{R}^{C\times C/r\times 1\times 1}$ are the 1×1 convolution layers. $F \in \mathbb{R}^{C/r\times(H+W)\times 1}$, r is the reduction ratio, δ is the nonlinear activation function. $F^H \in \mathbb{R}^{C/r\times H\times 1}$, $F^W \in \mathbb{R}^{C/r\times W\times 1}$ are obtained by splitting $F \in \mathbb{R}^{C/r\times(H+W)\times 1}$ along the spatial dimension. $\sigma(\cdot)$ denotes the sigmoid function. $\alpha^H \in \mathbb{R}^{C\times 1\times W}$ and $\alpha^W \in \mathbb{R}^{C\times H\times 1}$ are the learned spatial-spectral attention weights. Finally, the learned attention weights are used to adaptively recalibrate the input features X_a

$$\hat{X}_a = X_a \cdot \alpha^H \cdot \alpha^W. \tag{22}$$

III. Experimental Results and Analysis

A. Datasets and Evaluation Metrics

To verify the effectiveness of the proposed method, we compare our method with some other state-of-the art hyperspectral SR methods (including two-stream fusion network (TFNET) [29], MSDCNN [17], spatial-spectral residual network (SSRNET) [30], iterative regularization method based on tensor subspace representation (IR-TenSR) [31], nonnegative matrix factorization inspired deep unrolling network (NMF-DuNet) [32], and hyperspectral image super-resolution network (HSRNET) [20]) on various hyperspectral datasets, such as Columbia Imaging and Vision Laboratory (CAVE) [33], Harvard [34], Washington dc Mall (WDCM) [30], and real-world dataset (World View-2, WV2[1]). The CAVE dataset contains 32 indoor HSIs with size $512 \times 512 \times 31$, where the wavelength range is 400–700 nm. We select the first 20 images for training and the remaining 12 images for testing. The training patch size is set to $64 \times 64 \times 31$ with the cropping interval be 16. In the experiment, the LR-HSI is acquired from the HR-HSI images by applying an 8×8 Gaussian filter with a mean of 0 and a standard deviation of 2, and then downsampling the blurred HR-HSI images with a factor of 8. The HR-MSI is generated based on the spectral response of the IKONOS camera. The Harvard dataset has 50 HSIs, where the spatial size is 1040×1392 with 31 spectral bands. In the experiment, we first select 30 HSIs as training dataset and the remaining 20 images are used for testing. In the training process, the patch size is set to $64 \times 64 \times 31$ with a stride of 32. The corresponding LR-HSI and HR-MSI are generated by applying the same method as CAVE dataset. The WDCM dataset is obtained by the hyperspectral digital imagery collection experiment (HYDICE) over the WDCM. In this dataset, there are 210 bands covering the wavelength range from 0.4 to 2.5 um. After removing the bands with water absorption, the remaining 191 bands are adopted in the article, and each band has a size of 1280×307. In the experiment, the subregion $128 \times 128 \times 191$ is cropped for testing, and the rest region is chosen as the training set. During the training stage, the training patch size is set to $64 \times 64 \times 191$ with a stride of 16. Specifically, we zero-pad the test region of the WDCM to ensure that the test subregion and the remaining training region are nonoverlapping. Similarly, we also adopt the above method to get the corresponding LR-HSI dataset. The HR-MSI is simulated by choosing three bands from the remaining 191 bands (R-182, G-55, B-20). In addition, the generated LR-HSI and HR-MSI are added with Gaussian noise with different signal-to-noise ratios (SNR = 20, 25, and 30 dB) to further evaluate the effectiveness of the proposed CMSCN. The World View-2 dataset has a pair of real LR-MSI with size $418 \times 658 \times 8$ and a HR-red-green-blue (RGB) with size $1672 \times 2632 \times 3$. In the experiment, the lower right corner of the HR-RGB ($256 \times 256 \times 3$) and the LR-MSI ($64 \times 64 \times 8$) are cropped for testing, and the rest part is used as the training images. Since the reference HR-MSI is not available, we use the above method to produce the training set. In the training stage, the HR-RGB patches with the size of $64 \times 64 \times 3$ and the LR-MSI patches with size of $16 \times 16 \times 8$ are created for training.

Five widely used evaluation indexes are adopted, i.e., root mean square error (RMSE), peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) [35], erreur relative globale adimensionnelle de synthse (ERGAS) [36], and spectral angle mapper (SAM) [37]. RMSE and PSNR are often used to measure the difference between the restoration image and ground truth. The PSNR is defined as the average value of all spectral bands. SSIM reflects the spatial structure similarity between two images, which is also the average of all spectral bands. ERGAS represents the global statistical quality of the fused image. SAM is used to evaluate the spectral angle between the target and reference image. For RMSE, ERGAS, and SAM, the smaller the index, the better the fusion result. In contrast, the higher the PSNR and SSIM, the better the image quality. The ideal values for RMSE,

[1]https://www.l3harrisgeospatial.com/Data-Imagery/Satellite-Imagery/High-Resolution/WorldView-2

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TABLE I

ABLATION EXPERIMENTAL RESULTS OF THE PROPOSED CMSCN WITH OR WITHOUT NONLOCAL MODULE ON THE CAVE DATASET. THE BEST RESULTS ARE IN BOLD

Method	PSNR	SAM	ERGAS
w/o-NL	47.43	3.7522	0.6507
with-NL	47.44	3.7433	0.6519
CMSCN	**47.61**	**3.7418**	**0.6437**

CAVE [33], PSNR [35], (SAM) [37], (ERGAS) [36], w/o-NL [21], with-NL [21].

TABLE II

ABLATION EXPERIMENTAL RESULTS OF THE PROPOSED CMSCN WITH OR WITHOUT CSSC ON THE CAVE DATASET. THE BEST RESULTS ARE IN BOLD

Method	PSNR	SAM	ERGAS
w/o-CSSC	46.24	3.8594	0.7664
w/o-CS	46.94	3.8713	0.6949
CMSCN	**47.61**	**3.7418**	**0.6437**

PSNR, SSIM, ERGAS, and SAM are 0, $+\infty$, 1, 0, and 0, respectively.

B. Implementation Details

In each min-batch, 32 patches are randomly extracted as the input of the model. Adam optimizer [38] is selected for model training. The learning rate is initialized to 10^{-5} and reduced to half every 200 epochs. We implement our model by utilizing PyTorch with an Nvidia 2080Ti GPU. In the proposed network, we set all convolution layers to have 64 kernels of size 3×3 followed by a LeakyReLU [39] except for other instructions. We apply zero-padding strategy to retain the size of feature maps. The reduction radio in the spatial-spectral attention mechanism is set to 8. The number of DSCB and USCB in the network are set to 3, respectively. For upscale module, we use a 10×10 deconvolution layer [40] with a stride 8 to upsample the feature by a factor 8. $4\times$ upsampling use a 6×6 deconvolution layer with stride of 4. $2\times$ upsampling use a 4×4 deconvolution layer with stride of 2.

C. Ablation Analysis

In this section, we investigate the effects of different components, including CMNL, cross-scale self-calibrated structure (CSSC), CSs, and ESSA. During the training process, the proposed CMSCN and its variants share the same training settings.

Table I gives the quantitative results of ablation experiments on CAVE dataset. "w/o-NL" denotes the model of the proposed CMSCN without any NL. "With-NL" denotes the proposed CMSCN with NL. That is, in NL, the LR-HSI features are the only input of the NL to learn long-rang spatial information in LR-HSI and improve LR-HSI′s feature representation. CMSCN denotes the proposed CMSCN with CMNL. As shown in Table I, our CMSCN achieves the best performance in comparison with other variant methods. When removing NL, the performance of the network will be degraded. In previous studies, HR-MSI and the up-sampled LR-HSI are usually concatenated as the input of the network to learn the desired hyperspectral SR image. After integrating a nonlocal operation on the LR-HSI, the performance can be improved to a certain extent. In comparison with these models, we use CMNL to enhance the feature representation of LR-HSI by injecting HR spatial information from HR-MSI into LR-HSI before feeding them into the network. In Table I, the ablation results show that CMNL can help the network to gain more informative feature information and achieve a considerable performance improvement.

TABLE III

ABLATION EXPERIMENTAL RESULTS OF THE PROPOSED CMSCN WITH OR WITHOUT ESSA MECHANISM ON THE CAVE DATASET. THE BEST RESULTS ARE IN BOLD

Method	PSNR	SAM	ERGAS
With-CA	47.31	3.7524	0.6556
CMSCN	**47.61**	**3.7418**	**0.6437**

with-CA [27].

TABLE IV

AVERAGE COMPARISON RESULTS OF THE PROPOSED CMSCN AND OTHER STATE-OF-THE-ART METHODS ON THE CAVE DATASET. THE BEST RESULTS ARE IN BOLD

Methods	CAVE				
	PSNR	SAM	SSIM	RMSE	ERGAS
MSDCNN	46.10	4.0978	0.9920	0.0061	0.7816
SSRNET	45.20	5.1295	0.9893	0.0068	0.8809
IR-TenSR	43.89	5.0389	0.9854	0.0075	0.9646
NMF-DuNet	40.32	8.0115	0.9703	0.0132	1.6154
HSRNET	46.56	3.9141	0.9932	0.0057	0.7438
CMSCN	**47.61**	**3.7417**	**0.9933**	**0.0051**	**0.6437**

To demonstrate the effectiveness of the CSSC, the variant replaces the CSSC with the six residual blocks. Specifically, "w/o-CSSC" means that we use six residual blocks instead of CSSC in the network, "w/o-CS" denotes the model of CMSCN without the CSs. In Table II, the experimental results indicate that the CSSC structure can better restore the feature information in comparison with the traditional residual structure. After removing the CSs, the experimental indicators show different degradation, which proves the effectiveness of the CSs for feature extraction.

In the experiment, we also assess the effect of the attention mechanism. In Table III, "with-CA" denotes the model of CMSCN with common CA instead of ESSA mechanism. As shown in Table III, our CMSCN can achieve better reconstruction results than the variant model, which shows that the proposed ESSA can extract more important spatial-spectral feature information. The above comparisons consistently demonstrate the superiority of the proposed CMSCN.

D. Comparison With State-of-the-Art Methods

1) Results on CAVE Dataset. Table IV shows the quantitative results of our CMSCN and other state-of-the-art methods on the testing images. In comparison with these hyperspectral SR methods, the proposed CMSCN achieves the best performance on all five evaluation metrics. Further, we show the

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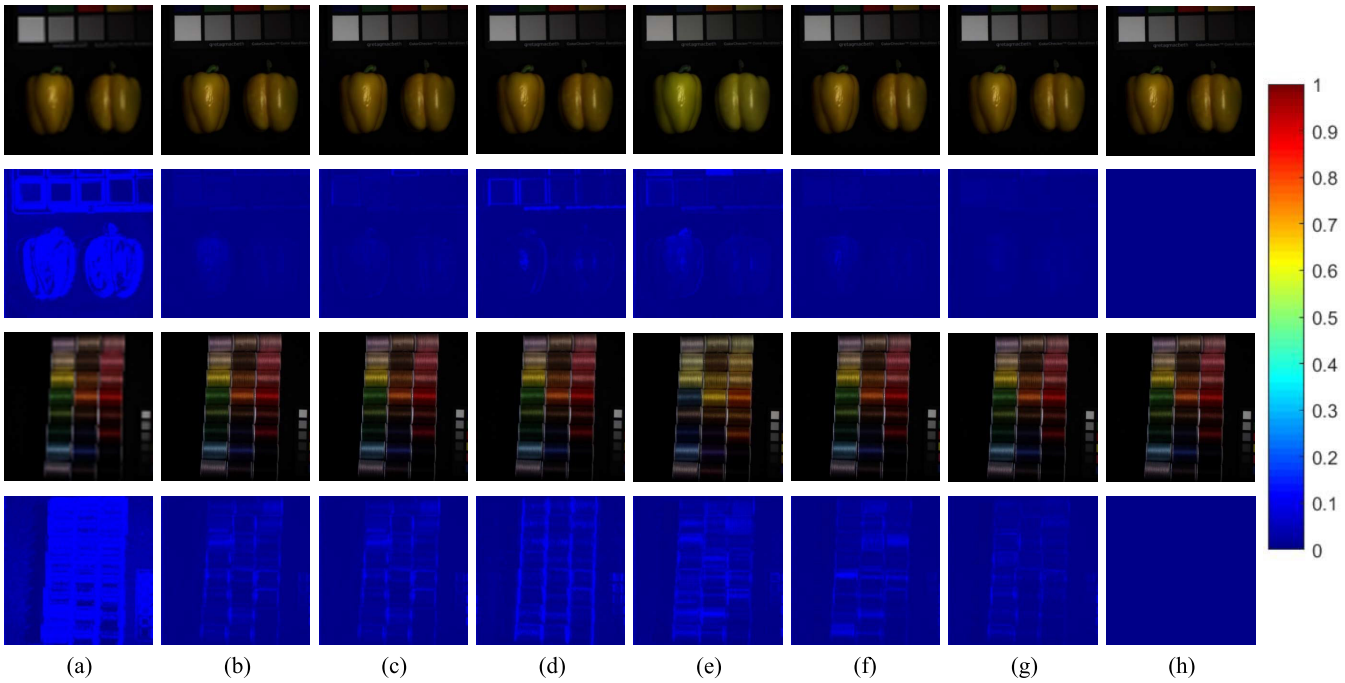


Fig. 6. Visual results of different methods on Cave dataset (i.e., $real_and_fake_peppers$, $thread_spools$). The first and third rows show the pseudocolor images (R-24, G-16, B-8) of the reconstruction HR-HSIs, and the second and fourth rows show the error maps between the fusion pseudocolor images and the corresponding ground-truth pseudocolor images. (a) Up-sampled LR-HSI. (b) MSDCNN. (c) SSRNET. (d) IR-TenSR. (e) NMF-DuNet. (f) HSRNET. (g) CMSCN. (h) ground truth image (GT).

TABLE V

AVERAGE COMPARISON RESULTS OF THE PROPOSED CMSCN AND OTHER STATE-OF-THE-ART METHODS ON $real_and_fake_peppers$ AND $thread_spools$ OF THE CAVE DATASET. THE BEST RESULTS ARE IN BOLD

Dataset	Methods	PSNR	SAM	SSIM	RMSE	ERGAS
$real_and_fake_peppers$	MSDCNN	53.23	2.7887	0.9964	0.0024	0.3978
	SSRNET	50.75	3.4690	0.9929	0.0034	0.5904
	IR-TenSR	48.79	3.1935	0.9938	0.0038	0.6470
	NMF-DuNet	45.52	6.7558	0.9804	0.0073	1.1058
	HSRNET	53.41	2.6601	0.9967	0.0023	0.3792
	CMSCN	**53.98**	**2.6094**	**0.9970**	**0.0022**	**0.3561**
$thread_spools$	MSDCNN	48.23	4.1696	0.9949	0.0046	0.6630
	SSRNET	47.16	5.1916	0.9927	0.0055	0.7843
	IR-TenSR	43.20	5.8978	0.9843	0.0082	1.2787
	NMF-DuNet	41.73	9.4035	0.9736	0.0107	1.6708
	HSRNET	47.80	4.0535	0.9953	0.0047	0.7013
	CMSCN	**49.68**	**3.8750**	**0.9958**	**0.0039**	**0.5714**

MSDCNN [17], SSRNET [30], IR-TenSR [31], NMF-DuNet [32], HSRNET [20].

TABLE VI

AVERAGE COMPARISON RESULTS OF THE PROPOSED CMSCN AND OTHER STATE-OF-THE-ART METHODS ON THE HARVARD DATASET. THE BEST RESULTS ARE IN BOLD

Methods	Harvard				
	PSNR	SAM	SSIM	RMSE	ERGAS
TFNET	42.30	5.1599	0.9686	0.0102	5.2768
MSDCNN	43.88	4.0755	0.9780	0.0085	2.7421
SSRNET	43.35	5.1042	0.9700	0.0092	5.4688
NMF-DuNet	38.97	6.0110	0.9494	0.0155	5.0554
HSRNET	43.86	4.0638	0.9791	0.0085	2.8216
CMSCN	**44.86**	**3.8862**	**0.9812**	**0.0079**	**1.9831**

detailed comparison of different methods on two images (i.e., $real_and_fake_peppers$ and $thread_spools$). In Table V, the experimental results show that the proposed CMSCN can obtain the comparable results and outperforms all the other methods. Notably, in Table V, on the $thread_spools$, the PSNR value of the proposed CMSCN over HSRNET is 1.88 dB, which benefits from the better reconstruction ability of our CMSCN.

In addition, we show visual results of different fusion methods with the pseudocolor images (R-24, G-16, B-8), which can be generated from the reconstruction HR-HSIs. In Fig. 6, it can be seen that our method yields best visual results among all existing compared methods. Take "$thread_spools$" for example, from the error results in Fig. 6, we can see that the Bicubic interpolation produces the worst visual result and heavy blurring artifacts. Later, although some deep learning methods can recover some structure details, their performance is still limited. The error maps of our method are lower than

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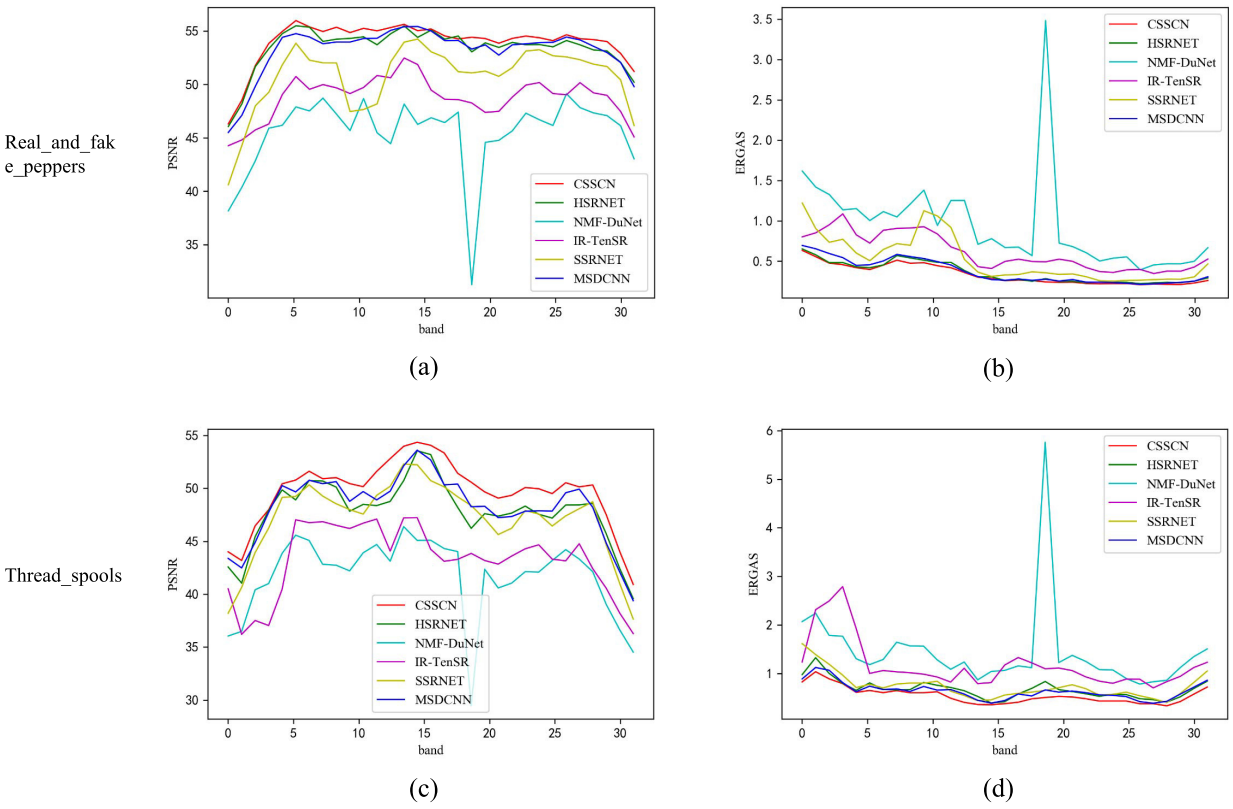


Fig. 7. Quantitative results of the proposed CMSCN with other state-of-the-art methods in terms of PSNR and ERGAS of each band of the reconstruction HR-HSIs on Cave dataset (i.e., (a) and (b): $real_and_fake_peppers$, (c) and (d): $thread_spools$).

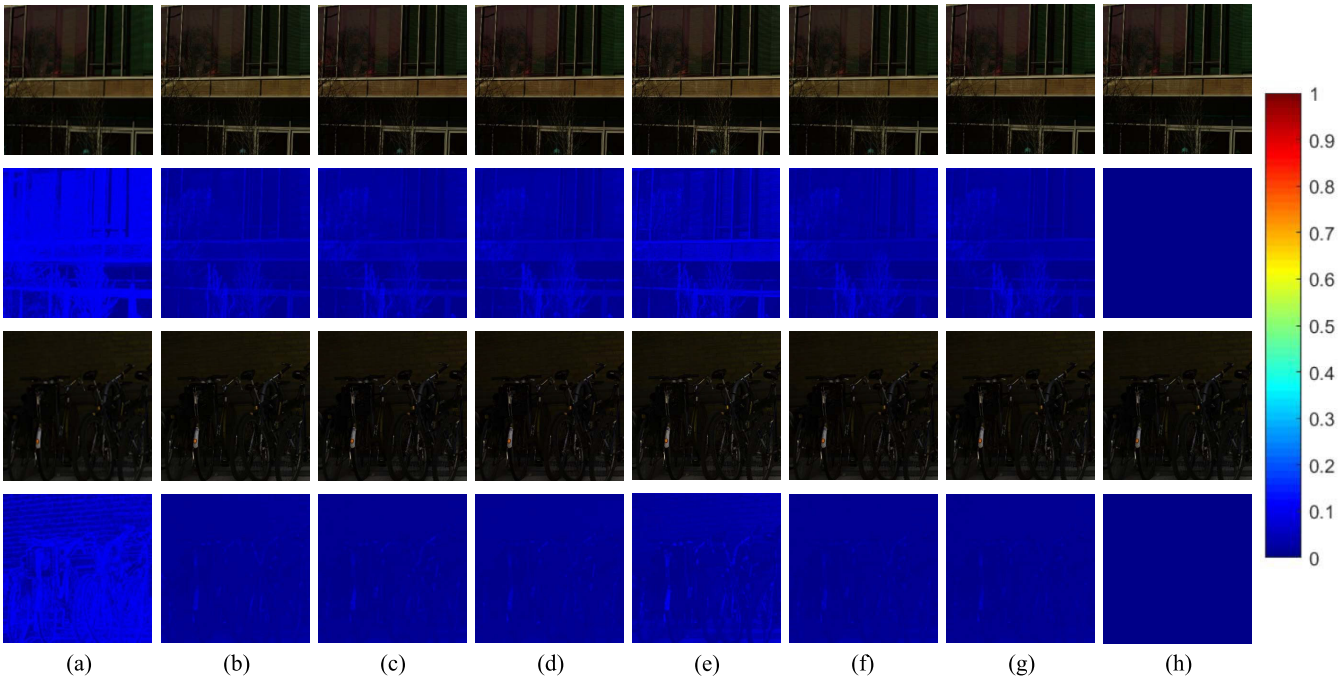


Fig. 8. Visual results of different methods on Harvard dataset (i.e., $imgf3$ and $imgf5$). The first and third rows show the pseudocolor images (R-26, G-20, B-10) of the reconstruction HR-HSIs, and the second and fourth rows show the error maps between the fusion pseudocolor images and the corresponding ground-truth pseudocolor images. (a) Up-sampled LR-HSI. (b) TFNET. (c) MSDCNN. (d) SSRNET. (e) NMF-DuNet. (f) HSRNET. (g) CMSCN. (h) GT.

other methods, which shows that our CMSCN has a good ability to reconstruct feature information. Fig. 7 illustrates the quantitative comparison of each spectral band provided by the different hyperspectral SR methods. It is worth noting that our

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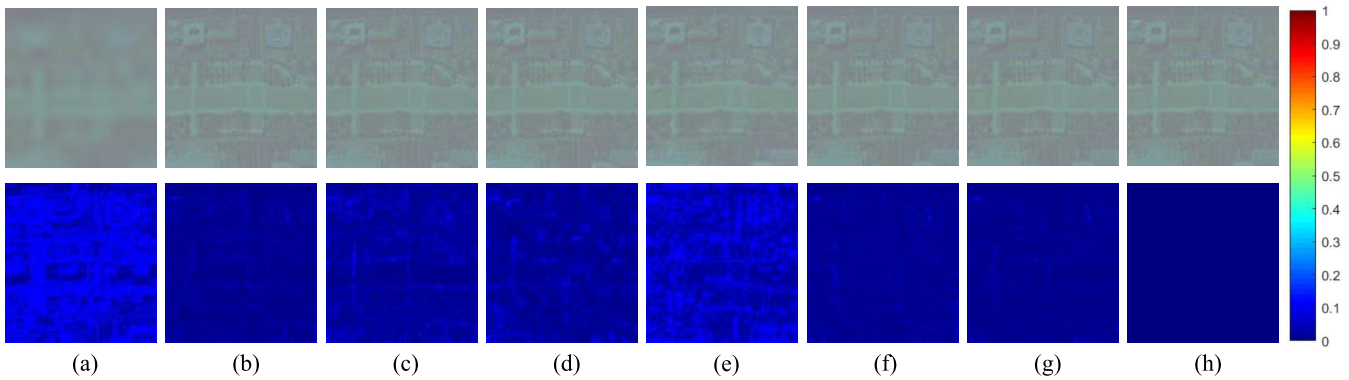


Fig. 9. Visual results of different methods on WDCM dataset. The first and third rows show the pseudocolor images (R-182, G-55, B-20, SNR=30 dB) of the reconstruction HR-HSIs, and the second and fourth rows show the error maps between the fusion pseudocolor images and the corresponding ground-truth pseudocolor images. (a) Up-sampled LR-HSI. (b) TFNET. (c) MSDCNN. (d) SSRNET. (e) NMF-DuNet. (f) HSRNET. (g) CMSCN. (h) GT.

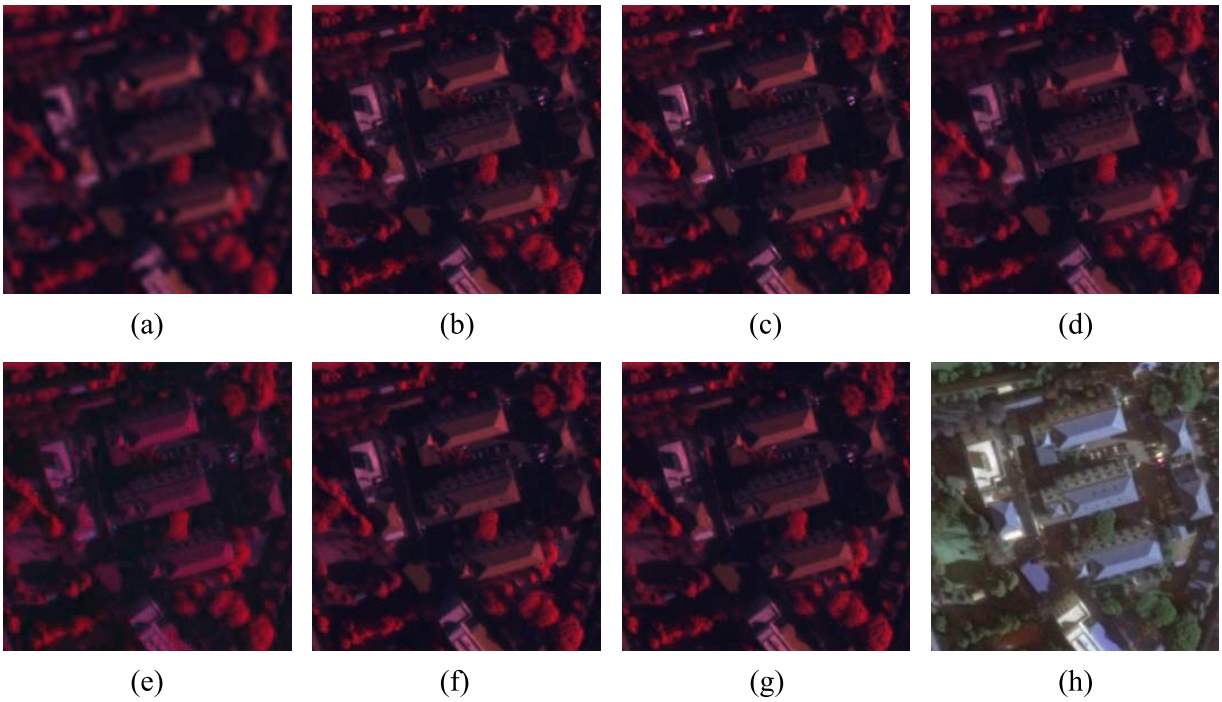


Fig. 10. Reconstruction results of World View-2 (R-6, G-3, B-2). (a) Up-sampled LR-MSI. (b) TFNET. (c) MSDCNN. (d) SSRNET. (e) NMF-DuNet. (f) HSRNET. (g) CMSCN. (h) RGB.

CMSCN method also obtains better reconstruction results in each spectral band in comparison with other state-of-the-art hyperspectral SR methods.

2) Results on Harvard Dataset. The experimental results of the proposed CMSCN and the comparative methods on the Harvard dataset are shown in Table VI. It is worthy to be remarked that we directly test the training model obtained from the CAVE dataset on the Harvard dataset without any refinement, which can reflect the generalization ability of these models. From Table VI, it can be seen that our CMSCN outperforms the compared methods in terms of all evaluation metrics. Table VII provides the detailed comparison for two images (i.e., $imgf3$ and $imgf5$) of the Harvard dataset. Specifically, in Table VII, for global statistical quality (ERGAS) or spectral preservation quality, the proposed CMSCN method always has a significant advantage over the second-best results.

For visual analysis from the Harvard dataset, we also show the error maps between the fusion pseudocolor images and the corresponding reference pseudocolor images (R-26, G-20, B-10). For the error maps in Fig. 8, it can be seen that our CMSCN method yields smallest error results over all the compared methods, which fully demonstrates the generalization ability of our method.

3) Results on WDCM Dataset. For the WDCM dataset, Table VIII lists the numerical evaluation results of different methods with different SNR (SNR $=$ 20, 25, and 30 dB). As can be seen, the proposed method can obtain superior

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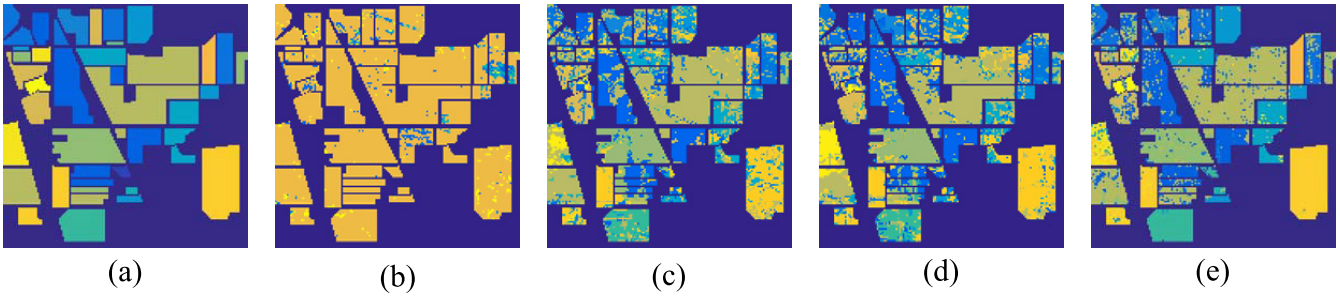


Fig. 11. Classification maps generated by SVM on different images of Indian Pines dataset. (a) Ground truth. (b) HR-MSI. (c) HR-HSI reconstructed by HSRNET. (d) HR-HSI reconstructed by CMSCN. (e) Ground truth Indian Pines dataset.

TABLE VII

AVERAGE COMPARISON RESULTS OF THE PROPOSED CMSCN AND OTHER STATE-OF-THE-ART METHODS ON $imgf3$ AND $imgf5$ OF THE HARVARD DATASET. THE BEST RESULTS ARE IN BOLD

Dataset	Methods	PSNR	SAM	SSIM	RMSE	ERGAS
$imgf3$	TFNET	40.65	6.9983	0.9575	0.0107	6.5313
	MSDCNN	41.91	6.0272	0.9706	0.0094	3.1809
	SSRNET	41.71	6.9541	0.9621	0.0091	5.2233
	NMF-DuNet	38.03	8.8016	0.9468	0.0160	5.4008
	HSRNET	42.14	5.7513	0.9733	0.0089	3.0438
	CMSCN	**43.16**	**5.4776**	**0.9769**	**0.0082**	**2.1295**
$imgf5$	TFNET	48.39	4.9101	0.9847	0.0041	3.6715
	MSDCNN	49.45	3.8566	0.9928	0.0035	2.0557
	SSRNET	48.45	4.8864	0.9844	0.0041	4.1126
	NMF-DuNet	45.17	5.6801	0.9836	0.0068	3.8691
	HSRNET	49.38	3.8991	0.9920	0.0035	2.1609
	CMSCN	**50.18**	**3.7338**	**0.9934**	**0.0033**	**1.6877**

TFNET [29], MSDCNN [17], SSRNET [30], NMF-DuNet [32], HSRNET [20], imgf 3 and imgf 5 [34].

TABLE VIII

AVERAGE COMPARISON RESULTS OF THE PROPOSED CMSCN AND OTHER STATE-OF-THE-ART METHODS ON THE WDCM DATASET WITH DIFFERENT SNRS. THE BEST RESULTS ARE IN BOLD

Methods	WDCM PSNR	SAM	SSIM	RMSE	ERGAS
SNR = 20 dB					
TFNET	50.47	0.4446	0.9844	0.0057	0.1246
MSDCNN	48.53	0.5352	0.9796	0.0065	0.1437
SSRNET	46.35	0.5371	0.9792	0.0069	0.1530
NMF-DuNet	39.79	0.7752	0.9487	0.0114	0.2622
HSRNET	50.13	0.4308	0.9850	0.0055	0.1210
CMSCN	**53.51**	**0.4121**	**0.9858**	**0.0054**	**0.1192**
SNR = 25 dB					
TFNET	51.05	0.3961	0.9878	0.0051	0.1124
MSDCNN	47.72	0.5480	0.9766	0.0068	0.1512
SSRNET	47.36	0.4762	0.9840	0.0061	0.1359
NMF-DuNet	40.29	0.7822	0.9555	0.0109	0.2489
HSRNET	49.93	0.4096	0.9868	0.0051	0.1128
CMSCN	**53.87**	**0.3798**	**0.9885**	**0.0050**	**0.1092**
SNR = 30 dB					
TFNET	51.78	0.3701	0.9895	0.0048	0.1050
MSDCNN	48.65	0.5023	0.9811	0.0062	0.1365
SSRNET	47.94	0.4663	0.9866	0.0057	0.1270
NMF-DuNet	40.50	0.7706	0.9560	0.0106	0.2434
HSRNET	50.42	0.3831	0.9888	0.0048	0.1056
CMSCN	**54.40**	**0.3329**	**0.9910**	**0.0043**	**0.0954**

performance on all metrics. Fig. 9 illustrates the difference maps of the fusion pseudocolor images and the corresponding reference pseudocolor images (R-182, G-55, B-20, SNR = 30 dB). In comparison with other methods, the reconstruction error map of our CMSCN method is smaller, which further validate the advantage of the proposed method over state-of-the-art methods.

4) Results on Real-World Dataset. To further evaluate the performance of the proposed method, we conduct a new experiment on the real-world dataset (World View-2). As shown in Fig. 10, in comparison with the other HSI SR methods, the proposed method can achieve better fusion performance and show stronger feature reconstruction ability.

E. Classification Performance

As discussed earlier, hyperspectral image SR is an image post-processing techniques that can produce a HR-HSI and extend the use of low-resolution image in high-level vision tasks [41], [42], [43]. To further demonstrate the effectiveness of the proposed method, we implement an experiment for HSI classification on Indian Pines dataset, including HR-MSI, HR-HSI recovered by HSRNET, HR-HSI recovered by the proposed CMSCN, and HR-HSI of the ground truth.

In the experiment, we use a support vector machine (SVM) with radial basis function (RBF) kernel for classification, and randomly select about 10% of the examples as the training set.

TABLE IX

CLASSIFICATION RESULTS OF SVM ON DIFFERENT IMAGES OF INDIAN PINES DATASET

Methods	OA	Kappa	AA
HR-MSI	0.2563	0.0337	0.0741
HSRNET	0.6146	0.5508	0.3966
CMSCN	0.6446	0.5887	0.4423
Ground Truth	0.7970	0.7671	0.6901

HSRNET [20].

Table IX shows the classification results of different methods in terms of average accuracy (AA), overall accuracy (OA), and Kappa coefficient (Kappa). As shown in Table IX, our CMSCN achieves the best classification performance on all quantitative metrics. In Fig. 11, we illustrate the classification maps of these methods on this dataset, which further confirm that our CMSCN has the ability to extract the important spatial-spectral features for hyperspectral image SR.

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IV. Conclusion

In this article, we propose an efficient CMSCN for hyperspectral and multispectral image fusion. Specifically, a CMNL is proposed to enhance the feature representation of LR-HSI by fusing the HR information of HR-MSI. In addition, we introduce a cross-scale self-calibrated convolution structure to fully learn spatial-spectral features at multiple scales, and make the model concentrate on more informative spatial-spectral feature information with the ESSA mechanism. The comprehensive experimental results show that our CMSCN method has better image fusion performance and robust generalization ability in comparison with the state-of-the-art methods.

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Huapeng Wu received the Ph.D. degree from the Nanjing University of Science and Technology, Nanjing, China, in 2021.

He is currently a Lecturer with the School of Computer Science, Nanjing Audit University, Nanjing. His research interests include hyperspectral image processing, super-resolution, and deep learning.



Jie Gui (Senior Member, IEEE) received the B.S. degree in computer science from Hohai University, Nanjing, China, in 2004, the M.S. degree in computer applied technology from the Hefei Institutes of Physical Science, Chinese Academy of Sciences, Hefei, China, in 2007, and the Ph.D. degree in pattern recognition and intelligent systems from the University of Science and Technology of China, Hefei, in 2010.

He is currently a Professor with the School of Cyber Science and Engineering, Southeast University, Nanjing. He has published more than 60 articles in international journals and conferences such as the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), the IEEE TRANSACTIONS ON CYBERNETICS (TCYB), the IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT), the IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS (TSMCS), ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), and ACM International Conference on Multimedia (ACM MM).

Dr. Gui is the Area Chair, the Senior Presiding Committee (PC) Member, or a PC Member of many conferences such as Neur information processing systems (IPS) and International Conference on Machine Learning (ICML).



Yang Xu (Member, IEEE) received the B.Sc. degree in applied mathematics and the Ph.D. degree in pattern recognition and intelligence systems from the Nanjing University of Science and Technology (NUST), Nanjing, China, in 2011 and 2016, respectively.

He is currently an Associate Professor with the School of Computer Science and Engineering, NUST. His research interests include hyperspectral image classification, hyperspectral detection, image processing, and machine learning.



Zebin Wu (Senior Member, IEEE) received the B.Sc. and Ph.D. degrees in computer science and technology from the Nanjing University of Science and Technology, Nanjing, China, in 2003 and 2007, respectively.

He is currently a Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology. He was a Visiting Scholar with the GIPSA-lab, Grenoble INP, University of Grenoble Alpes, Grenoble, France, from August 2018 to September 2018. He was a Visiting Scholar with the Department of Mathematics, University of California at Los Angeles, Los Angeles, CA, USA, from August 2016 to September 2016, and from July 2017 to August 2017. He was a Visiting Scholar with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politcnica, University of Extremadura, Ceres, Spain, from June 2014 to June 2015. His research interests include hyperspectral image processing, parallel computing, and big data processing.



Yuan Yan Tang (Life Fellow, IEEE) is currently the Director of Smart City Research Center, Zhuhai UM Science and Technology Research Institute, Zhuhai, China, is also the Emeritus Chair Professor with the University of Macau, Macau, China, and Hong Kong Baptist University, Hong Kong, an Adjunct Professor with Concordia University, Montreal, QC, Canada. His research interests include artificial intelligence, wavelets, pattern recognition, and image processing. He has published more than 600 academic articles and has authored or coauthored over 25 monographs, books, and book chapters. He is the Founder and the Editor-in-Chief of science citation index (SCI) journal *International Journal of Wavelets, Multiresolution and Information Processing* (IJWMIP).

Dr. Tang is an IEEE Life Fellow, IAPR Fellow, and Asia-Pacific Artificial Intelligence Association (AAIA) Fellow. He is the Founder and the General Chair of the series International Conferences on Wavelets Analysis and Pattern Recognition (ICWAPRs). He is the Founder and the Chair of the Macau Branch of International Associate of Pattern Recognition (IAPR). He has serviced as the General Chair, the Program Chair, and the Committee Member for many international conferences. He served as the Chairperson of 18th International Conference on Pattern Recognition (ICPR), which is the first time that the ICPR was hosted in China.



Zhihui Wei (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from South East University, Nanjing, China, in 1983, 1986, and 2003, respectively.

He is currently the Professor and the Doctoral Supervisor with the Nanjing University of Science and Technology, Nanjing. His research interests include mathematical image modeling, multiscale analysis, video and image coding and compressing, watermarking and steganography, speech and audio processing, mathematical methods, and machine learning techniques in data science, such as multiscale geometrical analysis, manifold learning and regularization, optimization method for big data processing, and deep learning.

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